

Metropolitan University, Sylhet
 Department of Computer Science and Engineering
 Spring Midterm Examination – 2024
 Program: B.Sc. in CSE; CSE(DAY)
 Batch: 58 (All section)
 Course: STA 215: Basic Statistics and Probability

Marks: 30

Time: 1.5 Hours

Answer any three questions. All parts of each question must be answered sequentially. Figures in the right margin indicate full marks.

1. a) What is a variable? Discuss different types of variables with examples. 2

b) Define with examples: i) Population ii) Sample 2

Imagine we're conducting a survey on the commute times of employees in a small town called Greenfield. We want to understand the distribution of commute times to help the town council plan better transportation infrastructure. We surveyed 50 randomly selected employees from various workplaces in Greenfield.

~~22, 18, 25, 30, 28, 35, 20, 27, 23, 40, 32, 26, 38, 42, 45, 21, 29, 31, 36, 24, 19, 33, 39, 43, 50, 22, 28, 34, 37, 48, 55, 60, 25, 22, 21, 26, 29, 33, 35, 38, 41, 45, 47, 52, 56, 65, 70, 75, 80, 90~~

c) Construct an exclusive frequency distribution choosing appropriate class interval. 2

d) Using the above data, draw a histogram and a frequency polygon on the same graph. 4

2. Rifat and Mahi play together in a basketball team. The list below shows the number of points that Rifat scored in each of 30 games:

29	19	28	30	13	21	23	15	34	24
29	17	43	12	24	25	41	19	26	40
45	23	21	32	37	24	18	15	24	36

a) Find the median and quartiles for these data. 3

b) Represent these data with a boxplot. 3

Mahi played in the same 30 games and his lowest and highest points total in a game were 19 and 41 respectively. The quartiles for Mahi were 27, 31 and 35 respectively.

c) Using the same scale draw a boxplot for Mahi's points total. 2

d) Compare and contrast the number of points scored per game by Rifat and Mahi. 2

class interval, fr.

24

3. The values of daily sales, to the nearest £, taken at a newsagent last year are summarized in the table below.

Sales	1-200	201-400	401-700	701-1000	1001-1500
Number of days	166	100	59	30	5

- a) Use interpolation to estimate the median and inter-quartile range of daily sales. 4
- b) Estimate the mean and standard deviation of these data. 4

The newsagent wants to compare last year's sales with other years.

- c) State whether the newsagent should use the median and the IQR or the mean and the standard deviation to compare daily sales. Give a reason for your answer. 2
4. a) What do you mean by statistics? Discuss the scope and limitations. 3

Mahi records the amount of rainfall, in mm, at his school each day for a week. The results are given below: 2.8, 5.6, 2.3, 9.4, 0.0, 0.5, 1.8

Rifat then records the amount of rainfall, x mm, at the school each day for the following 21 days. The results for the 21 days are summarized below: $\sum x = 84.6$

- b) Calculate the mean amount of rainfall during the whole 28 days. 2

Mahi realizes that he has transposed two of his figures. The number 5.6 should have been 6.5 and the number 2.3 should have been 3.2. Mahi corrects these figures.

- c) State, giving your reason, the effect this will have on the mean. 2

Imagine we have two companies, Tech Innovators Inc. (TII) and Green Energy Solutions (GES), with their annual return rates and standard deviations calculated from the past 5 years of stock performance as follows:

Tech Innovators Inc. (TII)
Mean Annual Return: 8%
Standard Deviation of Returns: 10%

Green Energy Solutions (GES)
Mean Annual Return: 5%
Standard Deviation of Returns: 7%

- d) Given the following information on two investment opportunities over the past 5 years, calculate the coefficient of variation (CV) for each company's stock returns and determine which investment is considered riskier based on the CV. 3

Metropolitan University, Sylhet
Department of Computer Science and Engineering
Spring Mid-Term Exam – 2024
Program: BSS in CSE
Batch: CSE 58 (E+G+H)

Course Title: Principles of Economics and Entrepreneurship Development
Course Code: GED-213

Time - 1 Hour and 30 Minutes

Full Marks: 30

[Answer any three sets of question. Figures in the right margin indicate full marks]

- ✓ 1 a) Briefly explain ten principles of Economics with hypothetical examples. 7
- b) How does the principle of "People face trade-offs" apply in the context of a student deciding between attending extracurricular activities and studying for exams? Provide a hypothetical scenario to illustrate this principle, and explain the trade-offs the student might encounter in making this decision. 3
- ✓ 2 a) List all the influences on buying plans that change demand, and for each influence, say whether it increases or decreases demand. 4
- b) Consider the market for smartphones, where the demand and supply functions are given by: Demand: $Q_d = 100 - 2P$ and Supply: $Q_s = 20 + 3P$
- Where Q_d represents the quantity demanded, Q_s represents the quantity supplied, and P represents the price of smartphones.
- I. Find the equilibrium price and quantity of smartphones. 6
- II. If the government imposes a price ceiling of \$30, what is the resulting quantity of smartphones exchanged in the market?
- 3 a) Explain cross elasticity of demand with a hypothetical example. 5
- b) What are the main influences on the elasticity of demand that change the demand? 5
- 4 a) Graphically explain consumer surplus and producer surplus 5
- b) In a perfectly competitive market, what are three possible short-run outcomes for a firm in response to changes in demand and cost conditions? 5

invisible

Metropolitan University, Sylhet
Department of Computer Science and Engineering
Midterm Examination - Spring 2024
Programme: B.Sc. in CSE (2nd Year 1st Semester); Batch: 58
Course: CSE 133: Data Structure

Time: 1 Hour 30 Mins

Marks: 30

Answer any **Three** sets of questions. All parts of each question must be answered sequentially. Figures in the right margin indicate full marks.

1. (a) Sort the following array in ascending order using **Quick Sort** algorithm and show every step. (5)
6, 5, 12, 10, 9, 1, 14, 8, 6, 7
- (b) Sort the same array in descending order using **Selection Sort** algorithm and show every step. (5)
2. (a) Which data structure is typically utilized in a text editor's undo feature? Justify your answer. (3)
- (b) Find 1101 using **Binary Search** from the following data and show each step. (5)
0000, 0001, 0010, 0011, 0100, 0101, 0110, 0111, 1000, 1001, 1010, 1011, 1100, 1101, 1110, 1111
- (c) What is the complexity of your provided solution? If the array is given in unsorted order and you have to apply **Merge Sort** followed by **Binary Search**, what will be the overall complexity? (2)
3. (a) Sort the following data in ascending order using **Merge Sort** algorithm and show every step. (5)
2 3 4 5 6
Mathematics, Geography, Programming, Microbiology, Python, Gastronomy, Mathemology, Musicology, Psychology
- (b) Suppose Metropolitan University authority decides to collect feedback from each department and assigns this task to the respective department heads. These heads then assign the task to batch supervisors, who gather feedback from students. The supervisors then send the feedback back to the department heads, who subsequently pass it on to the university authority. What process is being utilized to collect and share feedback in this scenario? Justify your answer. (3)
- (c) Which data structure would you utilize to maintain a line of customers for a restaurant management system? Justify your answer. (2)
4. (a) An empty **Queue** is given of size 2. Perform the following operations and show front and rear value at each step (Write overflow/underflow if they occur in any case). (4)
1. Dequeue () 2. Enqueue (8) 3. Enqueue (30) 4. Enqueue (40) 5. Dequeue ()
6. Dequeue ()
- (b) Convert the given **Infix** expression into a **Postfix** expression using a **Stack**, and show each step: $(A - 2 * (B + C) \wedge 3 / D * E) + F \wedge G$ $A \wedge B C + 30 / 2 \times \wedge \times - F G$ (3)
- (c) Evaluate the provided **Prefix** Expression using a **Stack**: $- + A * BC / D \wedge EF$ (3)
where, $A = 2, B = 3, C = 4, D = 16, E = 2, F = 3$.

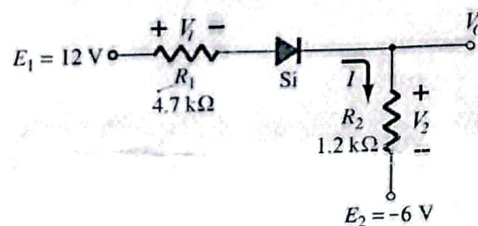
Metropolitan University, Sylhet
 Department of Computer Science and Engineering
 Mid-Term Examination, Spring-2024
 Program: B.Sc. in CSE; Batch: 58
 Course: CSE-131: Basic Electronics Engineering

Time: 1 Hour 30 Minutes

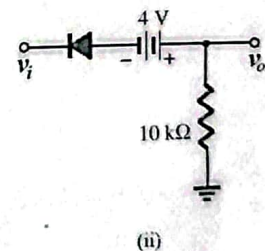
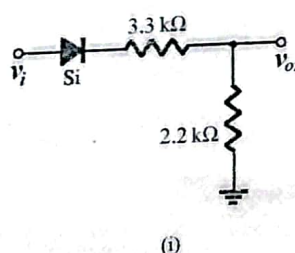
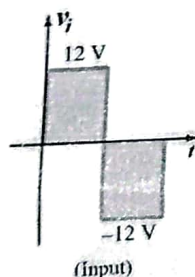
Answer any three (3) questions

Total Marks: 30

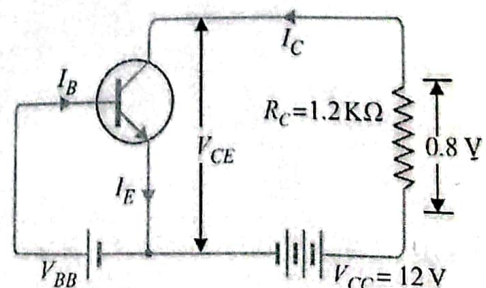
1. (a) Define intrinsic semiconductor and covalent bond in your own words. 2
- (b) Briefly explain the forward biasing condition of a $p-n$ junction diode using appropriate diagrams and notations. 4
- (c) A $p-n$ junction diode is operating at a temperature of 25°C and applied bias voltage is 0.9 V , the reverse saturation current at this stage is 60 nA , calculate following: 4
 - Thermal voltage of the diode.
 - Diode current through the diode if the ideality factor is 2.
2. (a) Sketch the $I-V$ characteristics curve of a zener diode and identify the key elements. 2
- (b) Does the doping process change the conductivity of a semiconductor material? If so, explain briefly using an energy band diagram how the doping process affects the conductivity. 4
- (c) Determine I , V_1 , V_2 and V_o in the circuit below. 4



3. (a) Define rectifier circuit and static resistance of a diode in your own words. 2
- (b) Draw a half wave rectifier circuit and explain its working principle with proper diagrams and waveforms. 4
- (c) Sketch the output waveform v_{o1} and v_{o2} for circuit in figure (i) and (ii) properly showing maximum and minimum voltages. Input v_i is the same for both circuits. 4



4. (a) Sketch the symbols of the nnp and pnp transistor and identify each terminal. 2
- (b) Explain the operation of a pnp transistor using appropriate diagrams and notations. 4
- (c) A transistor is connected in common emitter (CE) configuration in which collector supply is 12V and the voltage drop across resistance R_C connected in the collector circuit is 0.8V . The value of $R_C = 1.2\text{K}\Omega$. If $\sigma = 0.96$, determine following: 4
 - Collector-emitter voltage.
 - Base current.



Metropolitan University
Department of Computer Science & Engineering (CSE)

Mid-term Examination – Spring 2024

Program: CSE – Batch 58

Course Code: MAT- 135; Course Title: Matrices, Complex Variable and Fourier Analysis.

Time: 1 Hours 30 Minutes

Marks: 30

[Answer any three of the following question]

1. (a) Define the following matrices:

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- (i) Upper triangular matrix.
- (ii) Hermitial matrix.
- (iii) Nilpotent matrix.
- (iv) Involution matrix.

(b) Find the upper triangular matrix such that $A^3 = \begin{bmatrix} 8 & -57 \\ 0 & 27 \end{bmatrix}$.

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2. (a) If $A^{-1} = \begin{bmatrix} 1 & 2 & -4 \\ -1 & -1 & 5 \\ 2 & 7 & -3 \end{bmatrix}$; then find the original matrix A.

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(b) Solve the following linear equations with the help of the matrices:

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$$3x + 5y - 7z = 13$$

$$4x + y - 12z = 6$$

$$2x + 9y - 3z = 20$$

3. (a) Prove that every square matrix can be uniquely expressed as the sum of a symmetric matrix and skew-symmetric matrix.

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(b) Show that; $A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & -1 & 3 \\ 4 & 1 & 8 \end{bmatrix}$ and $B = \begin{bmatrix} -11 & 2 & 2 \\ -4 & 0 & 1 \\ 6 & -1 & -1 \end{bmatrix}$ are inverses.

$AB = BA = I$ 5

4. Consider the system:

$$x + 2y + z = 3$$

$$ay + 5z = 10$$

$$2x + 7y + az = b$$

(a) Find the values of a for which the system has a unique solution.

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(b) Find the pair of values (a, b) for which the system has more than one solution.

5

$A^2 = A$
 $A^2 = I$